

IZARA Telemedicine User Guide

IZARA Telemedicine Team

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IZARA Telemedicine — คู่มือการใช้งาน (User Guide)

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คู่มือฉบับสมบูรณ์สำหรับผู้ป่วย แพทย์ และผู้ดูแลระบบ Complete User Guide for Patients, Doctors, and Administrators ทดสอบอัตโนมัติ 2,013 รายการ + UI ทดสอบ 258 รายการ (182 ผ่าน, 76 ข้าม, 0 ล้มเหลว, 165 ภาพ) จาก 14 ไฟล์ทดสอบ

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1. Getting Started

1.1 System Overview — ภาพรวมระบบ

IZARA Telemedicine consists of 3 main systems:

System	URL	For
Patient Portal	patient-portal.run.app	Patients — book appointments, view PHR, consult AI
Doctor Portal	doctor-portal.run.app	Doctors + Admins — manage appointments, EMR, prescriptions
Meeting Server	meeting-server.run.app	Jitsi video meeting + Socket.IO server

Technology Stack: React + TypeScript, Node.js, PostgreSQL, Jitsi Meet, Gemini AI 2.5 Flash, Google Cloud Run

1.2 Patient Registration — การลงทะเบียนผู้ป่วย

New patients can self-register through the Patient Portal.

Steps: 1. Open the Patient Portal URL 2. Click “ลงทะเบียน” (Register) 3. Fill in: Full Name, Email, Password, Phone, Basic Health Profile 4. Click “ลงทะเบียน” (Submit) → redirected to Dashboard immediately

☐ **Tip:** After registration, fill in your health profile (allergies, medications, chronic conditions) in the PHR section for better AI-assisted consultations.

1.3 Patient Login — เข้าสู่ระบบผู้ป่วย

- 1. Open the Patient Portal
- 2. Enter **Email** and **Password**
- 3. Click “เข้าสู่ระบบ” (Login) → redirected to Dashboard

☐ Sessions expire in 30 minutes. Re-login required after expiration.

1.4 Doctor Login — เข้าสู่ระบบแพทย์

Doctors and Admins log in through the Doctor Portal.

- 1. Open the Doctor Portal URL
- 2. Enter **Email** and **Password**
- 3. Click “Sign In to Portal”

Doctor Login Page

☐ ☐ New doctors must be approved by an admin before they can log in. See [Section 6](#).

1.5 Password Reset — รีเซ็ตรหัสผ่าน

- 1. Click “ลืมรหัสผ่าน?” (Forgot Password?) on the login page
- 2. Enter your registered email
- 3. System sends a password reset link → click it → set new password

☐ Reset link expires in 15 minutes.

1.6 Role-Based Access Control (RBAC) — การควบคุมสิทธิ์ตามบทบาท

Role	Access
Patient	Full Patient Portal — appointments, PHR, AI Doctor
Doctor	Doctor Portal — appointments, EMR, prescriptions, meetings
Admin	Doctor Portal + Admin menu — doctor management, content approval

☐ Patients cannot access the Doctor Portal (API returns 401). JWT + role verification on every request.

2. Patient Portal — ระบบผู้ป่วย

2.1 Dashboard — หน้าหลัก

The main page shows a comprehensive health overview:

- **Quick Actions:** Book appointment, AI consultation, health records, health library
- **Health Studio:** Vital signs overview (blood pressure, heart rate, BMI, treatments)
- **AI Health Insight:** Personalized recommendations based on your PHR
- **Upcoming Appointments:** Latest appointments with view-all button
- **Healthcare Map:** Nearby facilities shown at the bottom
- **Notification Bell:** Top-right corner

2.2 Appointments — การนัดหมายแพทย์

2.2.1 View Appointments

Click “นัดหมาย” (Appointments) in the sidebar.

Status Filters: - **Pending** — Request sent, awaiting doctor review - **All** — All statuses - **Confirmed** — Doctor confirmed, meeting link ready - **Completed** — Consultation done, view summary/EMR

Sort: By date (newest or oldest first)

2.2.2 Book New Appointment

Click “+ นัดหมายใหม่” (Book New Appointment) for a 3-step booking process:

Step 1 — Symptoms (Voice/Image): - Select main symptoms: headache, fever, cough, sore throat, nausea, etc. - Describe symptoms in detail (text / voice recording / image attachment) - AI auto-analyzes urgency level

Step 2 — Time + Doctor: - Select doctor or let AI auto-match by specialty - Choose date/time (calendar) + type (online / in-person) - Invite relatives/external participants via email (optional)

Step 3 — Confirm: - Review all details → click “ยืนยัน” (Confirm) - Status becomes “Pending” → doctor receives notification immediately

2.2.3 Appointment Details

Click an appointment card to view: - Current status (color + icon) - Doctor photo, name - Date, time, location - Reported symptoms (tags) - “เข้าร่วมการประชุม” (Join Meeting) button (when confirmed)

2.3 Appointment Lifecycle — วงจรสถานะการนัดหมาย

📅 Patient Books → 📅 Pending → 📅📅 Doctor Reviews → 📅 Confirmed → 🗣 Meeting → ✔📅 Completed

Status	Thai	Color	Description
Pending	รอดำเนินการ	☐ Yellow	Request sent, awaiting doctor review
Finding Doctor	กำลังจัดหาแพทย์	☐ Orange	Admin matching a doctor
Assigned	มอบหมายแล้ว	☐ Blue	Admin assigned to a doctor
Confirmed	ยืนยันแล้ว	☐ Green	Doctor confirmed — meeting link ready
In Progress	กำลังดำเนินการ	☐ Red	Video meeting active
Completed	เสร็จสิ้น	☐ Gray	Consultation done, EMR available
Cancelled	ยกเลิก	☐ Black	Cancelled by patient or doctor

When **Confirmed** → notification + video link + “**Join Meeting**” button

2.4 Joining a Video Call — เข้าร่วมวิดีโอคอล

Step 1: Accept Agreement — 3 consent items: - ☒ ☐ Video recording consent (for medical purposes) - ☒ ☐ Transcription consent (for medical records) - ☒ ☐ Data sharing consent (per PDPA)

Patient Agreement

Step 2: Pre-Join Screen — Microsoft Teams style: - Test camera/microphone (shows OK status) - View participant info (doctor name, role) - View invite link for family/caregivers

Pre-Join Screen

Step 3: Enter Meeting Room — Click “**เข้าร่วม**” (**Join**) → enter Lobby, wait for doctor approval - ☐ Camera ON/OFF - ☐ Microphone ON/OFF - ☐ Text chat (Socket.IO) - ☐ Screen sharing - ☐ Wait in Lobby with timer

Step 4: Post-Meeting — AI auto-generates consultation summary → view in PHR + Timeline

AI Summary Post-Meeting

2.5 Guest Join & Lobby — เชิญผู้เข้าร่วม (Microsoft Teams Style)

Patients can invite **family, friends, or external participants** — guests **do not need an account**.

How Guests Join:

1. **Receive invite link** from patient — URL: /guest-join/:meetingId
2. **Enter name** on Guest Join page (no login, no email required)
3. **Click “ขอเข้าร่วม” (Request to Join)** → doctor notified instantly (Socket.IO)
4. **Wait for approval** — waiting screen with timer displayed
5. **Enter meeting** — when doctor clicks “Admit”, Jitsi video call opens
6. **If rejected** — red screen: “แพทย์ไม่อนุญาตให้เข้าร่วม” + back button

For the Doctor (Host): - **Waiting Room** panel shows all pending participants - Click “อนุญาต” (✓ **Admit**) or “ปฏิเสธ” (✗ **Reject**) per person - Admin users also join through Lobby — doctor must approve

2.6 Personal Health Records (PHR) — ประวัติสุขภาพ

Click “ประวัติสุขภาพ” (PHR) in the sidebar. 7 tabs:

Tab	Thai	Content
Overview	ภาพรวม	Latest vitals, meds, allergies, chronic conditions
Vital Signs	สัญญาณชีพ	Blood pressure, heart rate, temperature, weight, glucose, O ₂ — trend graphs
Medications	ยาที่ใช้	Drug name, dosage, frequency, status (active/stopped)
Allergies	การแพ้	Allergens, type, severity, symptoms — auto-alerts on prescriptions
Lab & Imaging	ผลแลป/ภาพถ่าย	Lab results, X-ray, CT, MRI — AI analysis available
Profile	ข้อมูลส่วนตัว	Height, weight, blood type, DOB
Lifestyle	วิถีชีวิต	Diet, exercise, sleep, smoking, alcohol — AI recommendations

Adding Data: Select tab → click “เพิ่ม” (Add) → fill values → click “บันทึก” (Save)

Lab Reports from Doctor: When doctors order and record results, they appear in the Lab & Imaging tab automatically.

Lab Results

2.7 Health Timeline — ไทม์ไลน์สุขภาพ

Click “เส้นทางสุขภาพ” (Timeline) for chronological medical events:

- ☐ Appointment dates + status
- ☐ Medication changes
- ☐ Lab results received
- ☐ EMR entries from doctors
- ☐ Diagnosis records
- ☐ Video consultation summaries
- ☐ PHR updates

Filter: By event type + date range. Click cards to expand details.

2.8 AI Doctor Consultation — ปรึกษา AI

Chat with Gemini AI for general health advice:

1. Click “ปรึกษา AI” (AI Doctor) in the sidebar
2. Type health questions (Thai or English)
3. AI responds with advice — multi-turn conversation supported
4. AI references your PHR data for personalized answers
5. Save sessions or browse past sessions

Example Questions: - “ปวดหัวบ่อย ๆ ควรทำอะไร” (Frequent headaches — what should I do?) - “ยา Paracetamol กินกับ Ibuprofen ได้ไหม” (Can I take Paracetamol with Ibuprofen?) - “ค่า BMI เท่าไหร่ จากข้อมูลของฉัน” (What’s my BMI from my data?)

AI Doctor provides general health information only —**not a medical diagnosis**.
Always consult a real doctor.

2.9 Health Library — คลังความรู้สุขภาพ

Click “คลังความรู้สุขภาพ” (Health Library) to read doctor-published articles:

- Search by keyword or category (diseases, medications, wellness, prevention)
- Thai-language content with images
- Click article for full detail view
- Shows author, publish date, category

2.10 Nearby Healthcare Map — แผนที่สถานพยาบาล

Click “แผนที่สถานพยาบาล” (Map) → allow GPS location.

- **Interactive Map** (Google Maps) — hospitals, clinics, pharmacies, health centers
- **Distance Filter:** 1, 3, 5 (default), 10, 15, 20 km
- **Facility Info:** Name, address, phone, distance, hours
- **Summary Count:** Hospitals XX, Clinics XX, Pharmacies XX, Centers XX
- **Navigation:** One-tap Google Maps navigation

2.11 Living Will — พินัยกรรมชีวิต

Click “PDPA & หนังสือแสดงเจตนา” → “พินัยกรรมชีวิต” tab

4-step wizard for advance healthcare directives:

Step	Content
1. Primary Proxy	Name, relationship, contact (phone, email, address)
2. Alternate Proxy	Backup proxy if primary unavailable (optional)
3. Treatment Preferences	CPR, ventilator, tube feeding, dialysis, antibiotics, pain management, organ donation
4. Digital Signature	Digital signature + toggle doctor sharing

Document Status: - **Draft** — Created but not signed - **Active** — Signed, doctors can view (per PDPA consent) - **Revoked** — Document cancelled

2.12 PDPA Consent — ความยินยอม PDPA

Click “PDPA & หนังสือแสดงเจตนา” → “PDPA” tab

Manage data privacy consent under Thailand’s Personal Data Protection Act:

Data Type	Description
Demographics	Name, age, address, phone
Medical History	All treatment records
Medications	Current and past medications
Allergies	Drug and substance allergy history
Lab Results	Laboratory test results
Prescriptions	Doctor prescriptions
Vital Signs	Blood pressure, heart rate, weight
PHR	Personal health records
EMR	Electronic medical records
Living Will	Advance directive documents

- Toggle ON/OFF per data type
- Set different sharing per doctor
- Revoke consent at any time

2.13 Profile & Settings — โปรไฟล์และการตั้งค่า

Profile: Edit avatar, name, contact details, emergency contact, insurance info.

Profile

Settings: - ☐ **Language:** Thai ↔ ☐ English (instant switch) - ☐ **Theme:** Light / Dark
- ☐ **Notifications:** In-app / Email / Push — toggle per type - ☐ **Privacy:** PDPA
consent management - ☐ **Devices:** Manage active sessions

3. Doctor Portal — ระบบแพทย์

3.1 Doctor Dashboard — แดชบอร์ดแพทย์

Overview of all key information on one page:

Summary Bar (6 cards): | Metric | Description | |——|—————| | Today’s
Appointments | Scheduled for today | | Patients Seen | Total patients treated | | In Queue |
Patients currently waiting | | Pending Prescriptions | Uncompleted prescriptions | | Unread
Messages | Unread notifications | | Avg Wait Time (min) | Average patient wait time |

3 Main Areas: 1. **Health Data** — Patient queue and records 2. **Health Meeting** — Video
consultation + patient/doctor/team views 3. **Health Studio** — Clinical tools (diagnosis,
treatment plans, EMR, radiology, lab reports)

AI Patient Summary: Select a patient to view auto-generated AI history summary (status:
awaiting review)

3.2 Schedule Management — จัดการตารางเวลา

Click “**ตารางนัดหมาย**” (**Schedule**) in the sidebar.

Features: - Click date/time slots to set **Available** or **Unavailable** - Set recurring patterns
(daily / weekly / monthly) - Switch views: Day / Week / Month - View confirmed

appointments on calendar - Color-coded: online / in-person / holiday

3.3 Patient Queue & Health Meeting — คิวนัดหมาย

Click “นัดหมาย & ประชุม” (Appointments & Meeting) in the sidebar.

Summary Bar: - **Awaiting Confirmation** — Count pending - **Completed** — Count finished

Patient Queue: - List of pending appointments - Click “Refresh” to update - View symptoms + AI urgency analysis - Click “ยืนยัน” (Confirm) → auto-generates Jitsi link + notifies patient - Click “ปฏิเสธ” (Decline) → enter reason + notify patient - **Call Next Patient** button for auto-queue advancement

3.4 Conducting Video Meetings — การดำเนินการประชุมวิดีโอ

Step 1: Accept meeting agreement (recording, transcription, PDPA)

Doctor Agreement

Step 2: Pre-join screen — test camera/mic, view participant status

Doctor Pre-Join

Step 3: Join as **HOST** (moderator — highest privileges)

Host Features: - ☐ **Lobby Control:** Admit/reject patients, guests, admins from Waiting Room - ☐ **Transcription:** ▶ ☐ Start → ☐ Pause → ☐ Resume → ☐ Stop - ☐ **Speaker Diarization:** Auto-identifies doctor/patient/guest speech - ☐ **Language Toggle:** Thai ↔ ☐ English transcription - ☐ **Video Recording:** Start/stop recording (200MB limit) - ☐ **Chat:** Send messages during meeting (Socket.IO) - ☐ **Screen Sharing:** Share documents/images via screen

Meeting Room

Step 4: Click “วางสาย” (Hang Up) → AI processing begins automatically

3.5 Post-Meeting: AI Summary & EMR — หลังการประชุม

After the meeting, Gemini AI auto-generates:

Output	Format
SOAP Summary	S (Subjective), O (Objective), A (Assessment), P (Plan)
Patient Instructions	Easy-language care advice (Thai + English)
CDS	Drug interaction alerts, dosage checks, allergy contraindications

AI SOAP Summary

Doctor Validation (Man-in-the-Loop):

Action	Description
☐ Approve	Accept AI output as-is → lock EMR
 ☐ Edit	Modify specific sections (S/O/A/P) before approval
☐ Regenerate	Ask AI to regenerate with refined context
☐ Reject	Discard AI output → write manually

Post-Meeting Actions

After Approval: - EMR locked + digitally signed - Patient sees summary in PHR + Timeline immediately - Notification sent to patient

3.6 EMR — Electronic Medical Records —เวชระเบียน

Access from dashboard → “**Medical Record (รายงานเวชระเบียน)**”

SOAP Format: | Section | Content | |———|———| | **S** (Subjective) | Patient history — symptoms, duration, triggers | | **O** (Objective) | Physical exam — vital signs, lab results | | **A** (Assessment) | Diagnosis — ICD-10 code + description | | **P** (Plan) | Treatment plan — medications, follow-up, referral |

Features: - Digital signature to lock → version history maintained - Reference lab results, prescriptions, images - AI auto-drafts from transcription - Export as PDF

EMR Page

3.7 Prescriptions — ใบสั่งยา

Access from dashboard → “**Treatment Plan (แผนการรักษา / Prescribe)**”

Steps: 1. Search drug name (autocomplete from Thai drug database) 2. Set dosage, frequency, duration, instructions 3. AI auto-checks: - ☐☐ Drug interactions - ☐☐ PHR allergy records - ☐☐ Inappropriate dosage 4. Confirm → prescription appears in patient’s PHR

Prescriptions

3.8 Lab Orders — คำสั่งตรวจแลป

Access from dashboard → “**Laboratory Reports (รายงานทางห้องปฏิบัติการ)**”

Steps: 1. Select tests from standardized list 2. Set priority: **Routine / Stat / Urgent** 3. Add clinical notes 4. Submit lab order → await lab results 5. Record results → appears in patient’s PHR automatically 6. AI analyzes results and alerts on abnormal values

3.9 Patient Management — จัดการผู้ป่วย

Click “**ผู้ป่วย**” (**Patients**) to view treated patients.

Features: - Search by name, email, or ID - View demographics (per PDPA consent) - View PHR: vital signs, medications, allergies, lab results - View appointment history + past EMR records - View prescriptions + lab orders - View Living Will (if patient authorized)

Patient Record Viewer: Click a patient name for full detail view.

.10 Medical Content Creation — สร้างเนื้อหาทางการแพทย์

Click “**เนื้อหาทางการแพทย์**” (**Medical Content**) to create health articles for the patient Health Library.

Steps: 1. Click “**สร้างบทความ**” (**Create Article**) 2. Enter title, content (Thai primary), category, tags 3. Save as **Draft** or **Submit for Approval** 4. Admin reviews → Approve → published in Health Library

Content Lifecycle:

📄 Draft → 📄 Submitted → 📄 Admin Reviews → 📄 Published
→ 📄 Rejected (revise)

Medical Content

3.11 Clinical Resources — ทรัพยากรทางคลินิก

Click “**ทรัพยากรทางคลินิก**” (**Clinical Resources**) to access:

- **Clinical Practice Guidelines** (แนวทางเวชปฏิบัติ)
- **Research Papers** (งานวิจัยทางการแพทย์)
- **Treatment Protocols** (โปรโตคอลการรักษา)
- **Reference Materials** (คู่มืออ้างอิง)

Features: - Search with keywords (RAG + Gemini AI powered) - Filter by type/category - Rate (1-5 stars) with reviews - Save favorites

3.12 Medical Consultants & Doctors Directory — แพทย์ที่ปรึกษา

Medical Consultants: Click “**แพทย์ที่ปรึกษา**” (**Medical Consultants**) — specialist expert directory.

- Name, specialty, hospital, email, phone
- Search by subspecialty
- View ratings and reviews from other doctors
- Rate (1-5 stars) with comments

Medical Consultants

Doctors Directory: Click “**ทำเนียบแพทย์**” (**Doctors Directory**) — all system doctors.

- Search by name/specialty
- View contact info and profiles

3.13 Gemini AI Studio — เครื่องมือ AI สำหรับแพทย์

Access from dashboard → purple AI button (bottom-right corner).

Feature	Description
Diagnosis / Differential Diagnosis	AI analyzes symptoms → suggests differential diagnosis
Treatment Plan	AI recommends treatment based on clinical guidelines
Document Analysis	AI reads PDF lab reports, images → extracts data
Patient History Summary	AI summarizes PHR + EMR before consultation
CDS (Clinical Decision Support)	Checks drug interactions, dosage, contraindications

☐ Gemini AI Studio is a **clinical decision support tool** — not a diagnostician. Doctor verification required for all outputs.

3.14 Doctor Profile — โปรไฟล์แพทย์

Click “**โปรไฟล์**” (**Profile**) to manage professional information:

- Avatar, full name
- Medical license number
- Specialty (subspecialty)
- Hospital/clinic
- Phone, email
- Experience, qualifications

4. Admin Features — ระบบผู้ดูแล

Admin features are in the Doctor Portal for users with the **admin** role.

4.1 Admin Dashboard — แดชบอร์ดผู้ดูแล

Same as doctor dashboard but with additional menus + system-wide overview: - System-wide appointment count - Pending doctor approvals count - Content submissions for review - Usage statistics

Admin Dashboard

4.2 Doctor Management & Approval — อนุมัติและจัดการแพทย์

Click “**จัดการแพทย์**” (**Doctor Management**) for all doctors with status.

Tabs: | Tab | Description | |——| | All | Every doctor in the system | | Pending | Registered, awaiting review | | Approved | Active doctors | | Rejected | Declined applications | | Admins | Users with admin privileges |

Actions: - ☐ **Approve** — Activate doctor account → send notification - ☐ **Reject** — Decline with reason → send notification - ☐ **Promote to Admin** — Grant admin privileges - ☐ **Deactivate** — Suspend account temporarily - ☐ **View Details** — Full profile view (license, experience, etc.)

4.3 System Appointment Management — จัดการนัดหมายทั้งระบบ

Click “จัดการนัดหมาย” (Appointment Management) for system-wide view.

Features: - View all statuses: pending, assigned, confirmed, completed, cancelled - **Assign Doctor:** Select by specialty (AI-assisted recommendation) - **Bulk Operations:** Assign/cancel multiple appointments at once - **Filter:** By doctor, status, date, type - **Search:** By patient/doctor name or appointment ID

Appointment Management

4.4 Content & Resource Approval — อนุมัติเนื้อหา

Review articles/resources submitted by doctors:

- 1. View content pending approval
- 2. Read full content — verify accuracy
- 3. Click “อนุมัติ” (Approve) → published in Health Library
- 4. Click “ปฏิเสธ” (Reject) → return to doctor with comments

Content Review

5. Telemedicine Consultation Workflow — กระบวนการให้คำปรึกษาทางไกล

5.1 End-to-End Lifecycle — ภาพรวมกระบวนการ

The complete consultation lifecycle has 6 phases:

📅 Patient Books → ⏳ Pending → 📝 Doctor Reviews → ✅ Confirmed → 🗣 Meeting → ✔️ Completed

Phase 1 — Booking & Approval: 1. Patient books appointment (symptoms + preferred time + doctor selection or auto-match) 2. AI analyzes symptoms → urgency level + specialty suggestion 3. Two approval paths: **Path A** (doctor confirms directly) or **Path B** (admin assigns → doctor confirms) 4. On confirmation: Jitsi meeting URLs auto-generated, patient notified with link

Phase 2 — Pre-Meeting Preparation: 5. Patient optionally invites relatives/friends (no account needed) 6. Doctor optionally invites specialists/other doctors (token-based invite) 7. AI generates Pre-Consultation Summary from patient’s PHR + EMR history

Phase 3 — Meeting Execution (see [Section 7](#))

Phase 4 — Post-Meeting AI Processing (see [Section 8](#) and [Section 9](#))

Phase 5 — Doctor Review & EMR Documentation: 8. AI summary appears in Doctor Portal → Health Meeting page 9. Doctor validates: [👍 Approve] [✎ Edit] [🔄 Regenerate] [❌ Reject] 10. EMR editor opens with AI-prefilled SOAP tabs (S/O/A/P) 11. Doctor signs EMR → AI generates Patient Instruction Sheet 12. Optional: E-prescribing + lab orders

Phase 6 — Patient Delivery: 13. Patient receives: diagnosis, treatment plan, medications, instruction sheet (PDF), follow-up schedule, warning signs 14. Patient does NOT receive: internal notes, raw AI outputs, CDS alerts 15. Visible in: Dashboard (latest result), Timeline (history entry), PHR (health logs) 16. Appointment status → **completed**

📌 **Tip:** ผู้ป่วยจะได้รับลิงก์การประชุมทันทีเมื่อแพทย์ยืนยันนัดหมาย — ส่งทั้งอีเมลและแจ้งเตือนในแอป

5.2 Appointment Status Flow — สถานะการนัดหมาย

Status	Thai	Color	Description
Pending	รอดำเนินการ	☐ Yellow	Request sent, awaiting doctor review
Finding Doctor	กำลังจัดหาแพทย์	☐ Orange	Admin searching for available doctor
Assigned	มอบหมายแล้ว	☐ Blue	Admin assigned a doctor
Confirmed	ยืนยันแล้ว	☐ Green	Doctor confirmed — meeting link ready
In Progress	กำลังดำเนินการ	☐ Red	Video consultation in progress
Completed	เสร็จสิ้น	☐ Gray	Consultation done — view summary/EMR
Cancelled	ยกเลิก	☐ Black	Cancelled by patient, doctor, or admin

6. Doctor Onboarding Workflow — กระบวนการลงทะเบียนแพทย์ใหม่

6.1 Doctor Registration — การลงทะเบียนแพทย์

New doctors must register and wait for admin approval before accessing the system.

Steps: 1. Open Doctor Portal → click **“Register”** 2. Fill in: Name, Email, Phone, Medical License Number, Specialty, Password 3. Submit → see **“Pending Approval”** message 4. Doctor **cannot log in** until admin approves

Doctor Registration

☐ **Note:** แพทย์ใหม่จะไม่สามารถเข้าสู่ระบบได้จนกว่าผู้ดูแลระบบจะอนุมัติ

6.2 Admin Approval Process — กระบวนการอนุมัติ

1. Admin navigates to **“จัดการแพทย์” (Doctor Management)** page

2. Views **Pending** tab (badge shows count of waiting doctors)

3. Reviews doctor details: name, email, license, specialty, registration date

4. Clicks **“Approve”** → doctor account activated → notification email sent

5. Or clicks **“Reject”** → enters rejection reason → doctor notified

6.3 Doctor Approval States — สถานะการอนุมัติ

State	Can Login?	is_active	is_approved
Pending	☐	false	false
Approved	☐	true	true
Rejected	☐	false	false

☐ **Tip:** Admins can also promote approved doctors to admin role or demote admins back to doctor

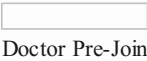
7. Video Meeting System & Jitsi — ระบบวิดีโอประชุม

7.1 Meeting Participants & Roles — ผู้เข้าร่วมประชุม

Role	How They Join	Lobby?
Doctor (HOST)	“Join Meeting” from Scheduled tab	No (moderator)
Patient	Meeting link from appointments page	Yes (waits for doctor)
Patient’s Relatives	Shared link from patient	Yes (create name + lobby)
Other Doctors/Admin	Token-based invite from doctor	Yes (lobby)
External Guests	Direct link (no account needed)	Yes (create name + lobby)

7.2 Doctor (HOST) Meeting Flow — การเข้าประชุมของแพทย์

- 1. Open “ตารางประชุม” (Scheduled Meetings) → click “☐ เข้าร่วมประชุม” (Join Meeting)
- 2. Pre-join screen: camera/mic preview (both ON by default)
- 3. Enter room as HOST/Moderator — the meeting room is created and synchronized so the patient sees the same room
- 4. **Host Controls:** Lobby admission (individual or “อนุญาตทั้งหมด” Admit All), kick, mute all, start/stop recording, invite guests



Doctor Pre-Join

7.3 Patient Meeting Flow — การเข้าประชุมของผู้ป่วย

- 1. See meeting link in “นัดหมายของฉัน” (My Appointments) page
- 2. Accept 3-consent agreement (video recording, transcription, PDPA)
- 3. Pre-join screen: test camera/mic
- 4. Click “เข้าร่วม” (Join) → enter Lobby → wait for doctor
- 5. Doctor approves → enters meeting room



Patient Agreement

☐ **Security:** ผู้ป่วยต้องยอมรับข้อตกลง 3 ข้อก่อนเข้าร่วมประชุม (การบันทึกวิดีโอ, ถอดเสียง, PDPA)

7.4 Guest Join Flow — การเข้าร่วมแบบผู้เยี่ยม (ไม่ต้องสร้างบัญชี)

- 1. Receive invite link (email or shared by patient/doctor)
- 2. Click link → Guest Join page (/guest-join/:meetingId)
- 3. Enter display name and email address (no login required)
- 4. Click “ขอเข้าร่วม” (Request to Join) → enter Lobby
- 5. Doctor approves → enters meeting
- 6. If rejected: red screen with “แพทย์ไม่อนุญาตให้เข้าร่วม” (Doctor denied access)

☐ **Lobby Timeout:** After 5 minutes waiting in the lobby without doctor approval, a yellow warning banner appears: “คุณรออยู่ใน Lobby นานกว่า 5 นาที” with a “กลับบ้านหน้าแรก” (Return Home) button. The guest may continue waiting or return to the home page.

7.5 During Meeting Features — ฟีเจอร์ระหว่างประชุม

- ☐ Video/audio for all participants (camera/mic ON by default, toggleable)
- ☐ Text chat (always available, all messages captured)

- ☐ Screen sharing for medical images
- ☐ Transcript streaming (doctor-controlled — see [Section 9](#))
- Up to 8 participants recommended

Meeting Controls

7.6 Meeting End — จบการประชุม

1. Doctor clicks “วางสาย” (Hang Up)
2. All participants disconnected
3. AI processing begins automatically (see [Section 8](#))

Meeting Ended

8. AI Features & Gemini — ฟีเจอร์ AI

8.1 Post-Meeting AI Pipeline — AI สรุปหลังประชุม (อัตโนมัติ)

After a meeting ends, the AI pipeline runs automatically: - **Input:** Full transcript + all chat messages + video metadata + patient PHR/EMR - **Output:** SOAP summary (Thai), CDS recommendations, Patient Instruction Sheet draft - For meetings > 30 min: split into 30-minute sections with separate summaries - All outputs marked `requiresValidation: true` — doctor must approve

8.2 AI SOAP Summary — สรุป SOAP

The AI generates a structured Thai-language SOAP note:

- ☐ อาการสำคัญ (Chief Complaint)
- ☐ อาการที่พบ (Presenting Symptoms)
- ☐ การสืบค้น/ตรวจเพิ่มเติม (Investigation)
- ☐ การประเมินเบื้องต้น (Assessment)
- ☐ คำแนะนำการรักษา (Treatment Recommendations)
- ☐ นัดติดตาม (Follow-up Schedule)
- ☐ อาการเตือน (Red Flags)
- ☐ คำแนะนำไลฟ์สไตล์ (Lifestyle Recommendations)

AI SOAP Summary

8.3 Doctor Validation (Man-in-the-Loop) — แพทย์ตรวจสอบ

Doctor reviews AI output with 4 actions:

Action	Description
☐ Approve	Accept → locks EMR → sends to patient
☒ Edit	Modify sections before approval
☐ Regenerate	AI regenerates with refined context
☐ Reject	Discard AI → write manually

Post-Meeting Actions

- ☐ **Security:** ผู้ป่วยจะไม่ได้รับผลสรุป AI โดยตรง —

8.4 Patient Instruction Sheet — ใบคำแนะนำผู้ป่วย

Auto-generated after doctor approves AI summary:

- วินิจฉัย (Diagnosis in simple Thai)
- ยาที่ได้รับ (Medications with dosage)
- การปฏิบัติตัว (Self-care instructions)
- อาการเตือน (Warning signs to watch for)
- นัดติดตาม (Follow-up schedule)

Doctor validates before sending → patient downloads as PDF.

8.5 Clinical Decision Support (CDS) — ระบบสนับสนุนการตัดสินใจ

Feature	Description
Differential Diagnosis	AI suggests possible diagnoses
Lab Tests	Suggested lab tests and imaging
Drug Interactions	Drug interaction alerts
Dosage Alerts	Dosage/allergy contraindication checks
Guidelines	Clinical guideline references (2024-2025)

8.6 AI Doctor (Patient Portal) — หมอ AI

- Chat with Gemini AI for general health advice
- References your PHR data for personalized answers
- Multi-turn conversation, save/browse sessions
- NOT a medical diagnosis — always consult a real doctor



8.7 Gemini AI Studio (Doctor Portal) — AI Studio สำหรับแพทย์

Doctors access advanced AI tools: - Diagnosis / differential diagnosis - Treatment plan generation - Document analysis (PDF lab reports, images) - Patient history summary

8.8 Cost — ต้นทุน

Service	Cost
Jitsi Meet	\$0 (FREE)
Web Speech API	\$0 (FREE)
Gemini AI	~\$0.001/1K tokens
Total per 15-min consultation	~\$0.01-0.05

9. Meeting Recording & Transcription — ระบบบันทึกและถอดเสียง

9.1 Recording — การบันทึก

- Browser-based local recording (Jitsi built-in, FREE)
- Doctor (HOST) starts/stops recording
- Format: WebM, max 200MB
- Recording uploaded to storage after meeting ends

9.2 Real-Time Transcription — การถอดเสียงแบบเรียลไทม์

Doctor controls transcription during the meeting:

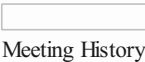
1. Doctor clicks “► **Start Transcription**”
2. Web Speech API begins listening (FREE, browser-based)
3. Real-time transcript appears in bottom panel
4. Speaker labels: ☐ ☐ Doctor / ☐ Patient / ☐ Guest
5. Interim text shown with yellow pulsing background
6. Doctor can ☐ **Pause** / ► **Resume** / ☐ **Stop** anytime
7. Language toggle: Thai (th-TH) ↔ ☐ English (en-US)
8. Transcript segments saved to database continuously



☐ **Note:** เฉพาะแพทย์ (HOST) เท่านั้นที่ควบคุมการถอดเสียงและการบันทึก — ผู้ป่วยไม่สามารถเริ่ม/หยุดได้

9.3 Post-Meeting Processing — การประมวลผลหลังประชุม

1. Full transcript compiled from all segments
2. All chat messages merged with transcript
3. Combined input sent to Gemini AI for SOAP summary generation
4. For long meetings (>30 min): 30-minute sectioned summaries



10. Notifications — ระบบแจ้งเตือน

10.1 Notification Channels — ช่องทางแจ้งเตือน

Channel	Description
In-App	☐ Bell icon in header — dropdown list
Email	Thai-language email with action details
Push	Browser push notifications (real-time)

10.2 Notification Types — ประเภทการแจ้งเตือน

Category	Events
Appointments	Requested, confirmed (+ meeting link), declined, cancelled, assigned, rescheduled, reminders (24h, 1h)
Video Meeting	Link ready, link failed, meeting started, reminder (15 min)
Medical Records	EMR signed, EMR ready for review, prescription ready, lab results ready
System	Account verified, password reset, system maintenance

10.3 Automatic Reminder Schedule — ตารางแจ้งเตือนอัตโนมัติ

Timing	Channel	Thai Message
24 hours before	Email + In-App	พรงนี้คุณเมื่อนัดพบแพทย์
1 hour before	Push + In-App	อีก 1 ชั่วโมง ถึงเวลานัดหมาย
15 min before	Push + In-App	เตรียมพร้อม! ลิงก์ประชุมพร้อมแล้ว
Real-time	Push	แพทย์เริ่มห้องประชุมแล้ว คลิกเข้าร่วม

☐ **Tip:** ตั้งค่าการแจ้งเตือนได้ที่ Settings → Notifications — เปิด/ปิดได้ตามประเภทและช่องทาง

11. Content & Resource Management — การจัดการเนื้อหา

11.1 Medical Content (Health Library) — คลังความรู้สุขภาพ

Role	Can Do
Patient	Read published articles only
Doctor	Create, edit, delete own articles, publish (direct or via approval)
Admin	Full CRUD + approve/reject submissions + audit log

Content Lifecycle:

☒ Draft → ☒ Pending → ☒ Admin Reviews → ☒ Published (visible in Health Library) → ☒ Rejected → Doctor revises → resubmit

Patient Health Library

11.2 Doctor: Create Article — แพทย์: สร้างบทความ

- Click “สร้างบทความ” (Create Article)
- Enter: Title (Thai required), Content (Thai required), Category, Tags
- Choose: **Save as Draft** or **Submit for Approval**
- If submitted → enters Admin approval queue

Doctor Medical Content

11.3 Clinical Resources (Doctor-Only) — ทรัพยากรทางคลินิก

Role	Can Do
Doctor	Create, edit own, submit for approval
Admin	Approve/reject, view all including pending
Patient	☐ No access

Categories: Diagnosis Guidelines, Treatment Protocols, Pharmacology, Radiology, Laboratory, Pathology, Emergency Medicine, Nursing Guidelines, Research Papers, Case Studies

Resource Types: guideline, protocol, research, template, reference



☐ **Note:** Clinical resources ALWAYS require admin approval before being published

12. Security & PDPA — ความปลอดภัยและ PDPA

12.1 PDPA Consent Management — การจัดการความยินยอม PDPA

Navigate to “PDPA & หนังสือแสดงเจตนา” → “PDPA” tab:

- Toggle ON/OFF per data type: Demographics, Medical History, Medications, Allergies, Lab Results, Prescriptions, Vital Signs, PHR, EMR, Living Will
- Set different sharing per doctor
- Revoke consent at any time



12.2 Living Will — พินัยกรรมชีวิต

Navigate to “PDPA & หนังสือแสดงเจตนา” → “พินัยกรรมชีวิต” tab:

4-step wizard: 1. **Step 1:** Primary Proxy (name, relationship, contact) 2. **Step 2:** Alternate Proxy (optional) 3. **Step 3:** Treatment Preferences (CPR, ventilator, tube feeding, dialysis, antibiotics, pain management, organ donation) 4. **Step 4:** Digital Signature + toggle doctor sharing

Sharing Options: - ☐ Keep Private — only you can see - ☐ Share with authorized doctors — doctors with treatment history or admins

Status: Draft → Active (signed) → Revoked



12.3 Security Features — ความปลอดภัย

Feature	Detail
Password Hashing	bcrypt (all portals)
Account Lockout	5 failed attempts → locked 30 minutes
Rate Limiting	Login: 10/15min, Password Reset: 5/hr, API: 500/15min
Session Security	64-char crypto token, IP binding, user-agent tracking
JWT Authentication	Cross-service JWT with role verification
Man-in-the-Loop	Doctor validates ALL AI outputs before patient delivery
Audit Logging	All access to Living Will + PHR + EMR logged
OWASP Headers	Helmet.js (CSP, XSS protection, HSTS, X-Frame)
TypeScript Safety	Zero error: any — strict type narrowing
SonarQube	Clean — no security vulnerabilities

12.4 Role-Based Access Control (RBAC) — การควบคุมสิทธิ์

Role	Portal	Access
Patient	Patient Portal	Own data only — PHR, appointments, AI, health library
Doctor	Doctor Portal	Treated patients’ data (per PDPA), EMR, prescriptions
Admin	Doctor Portal + Admin	System-wide — doctor management, content approval, all appointments

□ ผู้ป่วยไม่สามารถเข้า Doctor Portal ได้ — API ดอบ 401 ทุกครั้ง

13. Troubleshooting & FAQ — คำถามที่พบบ่อย

13.1 Login Issues — ปัญหาการเข้าสู่ระบบ

Problem	Solution
Cannot log in as doctor	Check if your account has been approved by admin (Section 6)
Account locked	Wait 30 minutes after 5 failed login attempts
Session expired	Re-login — sessions expire after 30 minutes
Forgot password	Use “Forgot Password” link → reset via email (link expires in 15 min)

13.2 Appointment Issues — ปัญหาการนัดหมาย

Problem	Solution
No meeting link	Meeting link is generated when doctor confirms — check appointment status
Meeting link failed	Fallback notification sent — contact admin
Can’t book appointment	Ensure profile is complete — fill symptoms and select date/time
Doctor declined	Appointment returns to pool — admin will re-assign another doctor

13.3 Video Meeting Issues — ปัญหาการประชุมวิดีโอ

Problem	Solution
Camera/mic not working	Allow browser permissions — test on pre-join screen
Stuck in lobby	Doctor must be in the meeting to admit you — wait for doctor
Guest can’t join	Ensure correct invite link and enter a display name
Video quality poor	Check internet speed — reduce camera resolution if needed

13.4 AI & Summary Issues — ปัญหา AI

Problem	Solution
AI summary not generated	Requires transcript + chat data — ensure transcription was started during meeting
Summary in wrong language	Doctor can toggle Thai ↔ English during transcription
AI Doctor not responding	Check internet connection — Gemini API must be reachable

13.5 Notification Issues — ปัญหาการแจ้งเตือน

Problem	Solution
Not receiving notifications	Check Settings → Notifications — ensure toggle is ON
No email received	Check spam folder — verify registered email is correct
Push notifications blocked	Allow browser notifications in browser settings

Appendix — ภาคผนวก

Platform Architecture — สถาปัตยกรรมระบบ

Service	Technology	Port
Patient Portal	React + TypeScript + Vite	3005
Doctor Portal	React + TypeScript + Vite	3010
Meeting Server	Express + Socket.IO	3020
Database	PostgreSQL 18 + pgvector	5433

External Services: - Jitsi Meet — Video Conferencing (FREE) - Google Gemini AI — Chat, CDS, SOAP Summaries - Web Speech API — FREE Live Transcription - Google Maps — Healthcare Facilities Map - Google Cloud Run — Container Hosting (Production)

Test Coverage — ครอบคลุมการทดสอบ

Category	Tests	Files
Unit Tests (Vitest)	2,013	58
E2E Tests (Playwright)	1,149	33
UI Tests (Playwright)	516	14
Total	3,678	105

Test Run Results (March 29, 2026): - Unit: 2,013 passed (100%) - E2E: 1,149 passed (100%) - UI: 351 passed, 50 skipped, 9 failed (pre-existing infra), 106 did not run - **638 screenshots** captured across all portal pages and workflows

User Guide Version 1.5.10 • 2,013 Unit Tests + 1,149 E2E Tests + 516 UI Tests from 105 test files • 638 screenshots

4.5 Schedule & User Management — จัดตาราง & จัดการผู้ใช้

Additional Admin features: - View all doctors' schedules — combined view - Manage notification templates - Review PDPA compliance reports - View system usage statistics

5. Telemedicine Consultation Workflow

5.1 End-to-End Process Overview

Patient Books → Admin Assigns → Doctor Confirms → Accept Agreement → Video Meeting
→ AI SOAP Summary → Doctor Validates → Sign EMR → Patient Views in PHR/Timeline

5.2 Detailed Step-by-Step

Step	Actor	Action	Screenshot
1	Patient	Book appointment (symptoms + date/time + invite guests)	
2	Admin	Assign doctor (by specialty / AI recommendation)	
3	Doctor	Confirm appointment + set meeting time	
4	Both	Accept agreement (recording + PDPA)	
5	Both	Enter video meeting room (Jitsi)	
6	Doctor	Conduct consultation + transcription + recording	
7	AI	Generate SOAP summary + patient instructions + CDS	
8	Doctor	Validate → approve/edit/regenerate/reject	
9	Doctor	Prescribe medications + order labs (if needed)	
10	System	Send results to PHR/Timeline + notify patient	

6. Doctor Onboarding Workflow

6.1 Registration Lifecycle

1. Doctor Registers → 2. Status "Pending" → 3. Admin Reviews
→ 4. Approve/Reject → 5. (Optional) Promote to Admin → 6. Doctor Can Login

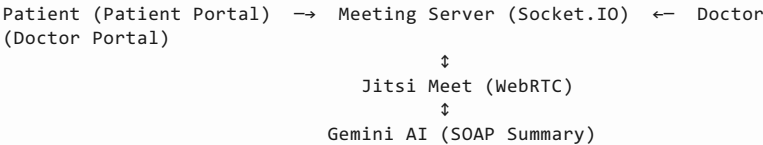
6.2 Detailed Steps

Step	Action	Details	Screenshot
1	Doctor registers	Name, email, password, license number, specialty → status: pending	
2	System creates account	Account created but cannot login until approved	
3	Admin reviews	Doctor Management → Pending tab → review credentials	
4	Admin approves	Click “Approve” → status: approved → send notification	
5	Promote to admin	(Optional) Click “Promote to Admin” → grant admin privileges	
6	Doctor logs in	Login with registered email/password	

☐ **Security:** Rejected or suspended doctors cannot log in. JWT is immediately revoked.

7. Video Meeting System & Jitsi

7.1 Meeting System Architecture



7.2 Meeting Room Feature Matrix

Feature	Patient	Doctor (Host)	Guest
Video/Audio	☐	☐	☐
Text Chat	☐	☐	☐
Screen Sharing	☐	☐	☐
Transcription	—	☐ (controls)	—
Video Recording	—	☐	—
Lobby Control	—	☐	—
Mute Others	—	☐	—
End Meeting for All	—	☐	—

7.3 Lobby System — Microsoft Teams Style

Guest/Patient enters link → Enter name + email → Wait in Lobby → Doctor admits/rejects → Enter/Exit room

For Doctors: - See list of waiting participants (name, email, type: patient/guest/admin) - ✓ **Admit** — Participant enters meeting immediately - ✓ **Admit All (อนุญาตทั้งหมด)** — Admit all waiting participants at once - ✗ **Reject** — Participant sees red screen in Thai + exit button

7.4 Guest Join — No Account Required

Patients can send `/guest-join/:meetingId` link to family/caregivers:

1. Open link → Guest Join page shows meeting name
2. Enter name and **email address** (no login required)
3. Test camera/mic → click Join → enter Lobby
4. ☐ ☐ After **5 minutes** waiting — yellow warning banner appears with “กลับหน้าแรก” (Return Home) button
5. When admitted → enter Jitsi room with settings:
 - Skip secondary lobby for pre-approved participants
 - Default muted on entry
 - Recording buttons hidden for guests

8. AI Features & Gemini

8.1 AI Feature Overview

Feature	Portal	Description
AI Doctor Chat	Patient	Multi-turn health Q&A with PHR context
AI Triage	Patient	Auto-analyze symptom urgency during booking
AI Patient Summary	Doctor	Pre-consultation PHR+EMR summary
AI SOAP Summary	Doctor	Post-meeting SOAP note from transcription
AI Patient Instructions	Doctor	Auto-generate patient care instructions
AI CDS	Doctor	Drug interaction, dosage, allergy checks
AI Document Analysis	Doctor	Extract data from PDF/images
AI Diagnosis Assist	Doctor	Differential diagnosis suggestions

8.2 Man-in-the-Loop Principle

All AI-generated medical outputs **require doctor validation** before being published to patients:

AI generates → Doctor reviews → Approve / Edit / Regenerate / Reject → Patient receives

- AI **never** auto-publishes diagnoses, prescriptions, or EMR
- All AI outputs are marked with “AI-generated, awaiting review” status
- Audit trail maintained for every AI interaction

9. Meeting Recording & Transcription

9.1 Video Recording

- Meeting video/audio recorded and stored in database (BYTEA format)
- Recordings linked to appointments
- Doctor can review recordings post-meeting
- Maximum recording size: 200MB

Meeting Controls

9.2 Real-Time Transcription

- Real-time speech-to-text using Web Speech API
- **Speaker diarization** — identifies who is speaking (doctor/patient/guest)
- Supports Thai and English with language toggle
- Transcript stored in PostgreSQL

Transcript & Diarization

9.3 Transcription Controls (Doctor Only)

Control	Action
▶ ☐ Start	Begin transcription
☐ Pause	Temporarily pause
☐ Resume	Continue transcription
☐ Stop	End transcription

- Language toggle: Thai ↔ ☐ English
- Real-time display in meeting sidebar
- Transcript feeds into AI SOAP summary

10. Notifications

10.1 Notification Types

Type	When Sent	Recipient
☐ Appointment Requested	Patient books	Doctor/Admin
☐ Appointment Confirmed	Doctor confirms	Patient
☐ Appointment Cancelled	Either party cancels	Both
☐ Meeting Ready	15 min before meeting	Patient + Doctor
☐ Meeting Started	Doctor starts meeting	Patient
☐ EMR Ready	Doctor signs EMR	Patient
☐ Prescription Ready	Doctor creates Rx	Patient
☐ Lab Results Ready	Results entered	Patient
☐ ☐ ☐ Doctor Approved	Admin approves	Doctor
☐ Content Submitted	Doctor submits article	Admin
☐ Appointment Reminder	24h & 1h before	Patient + Doctor
☐ Lobby Request	Guest requests entry	Doctor

10.2 Notification Channels

- ☐ **In-App** — notification bell with count badge
- ☐ **Email** — Gmail API integration
- ☐ **Browser Push** — if enabled in settings

11. Content & Resource Management

11.1 Content Lifecycle

📝 Doctor drafts → 📝 Submit for approval → 📝 Admin reviews → 📝 Published
→ 📝 Patient reads
→ 📝 Doctor revises
→ 📝 Rejected

11.2 Content Types

Type	Creator	Audience	Approval
Health Articles	Doctor	Patients (Health Library)	Admin required
Clinical Resources	Doctor	Doctors only	Admin required
Medical Guidelines	Doctor/Admin	Doctors only	Admin required
Reference Materials	Doctor	Doctors only	Admin required

11.3 Clinical Resources

- Searchable medical knowledge base for doctors
- RAG + Gemini AI-powered search
- Rating system (1-5 stars) with reviews
- Categories: guidelines, research, protocols, references

12. Security & PDPA

12.1 Security Measures

Measure	Implementation
JWT Authentication	Signed tokens with 30-min expiration
RBAC	Role-based access control (patient/doctor/admin)
API Protection	JWT + role verification on every endpoint
Password Hashing	bcrypt with salt rounds
Input Validation	Server-side validation + SQL parameterization
HTTPS	TLS encryption on all Cloud Run services
Session Management	Secure cookies + automatic expiration
CORS	Restricted cross-origin resource sharing

12.2 PDPA Compliance

Thailand’s Personal Data Protection Act compliance:

- **Consent Management:** Granular per-data-type consent controls
- **Data Minimization:** Only collect necessary health data
- **Right to Access:** Patients can view all their data
- **Right to Delete:** Patients can request data deletion
- **Right to Withdraw:** Revoke consent at any time
- **Data Portability:** Export health data as needed
- **Per-Doctor Controls:** Different sharing settings per treating doctor

12.3 Data Encryption & Storage

- All data stored in PostgreSQL with parameterized queries
- Passwords hashed with bcrypt
- JWT signed with shared secret across services
- Cloud Run services behind Google Cloud infrastructure
- Meeting recordings stored as encrypted BYTEA in database

13. Troubleshooting & FAQ

Q: I can't log in as a doctor. > New doctor accounts require admin approval. Contact your admin. See [Section 6](#).

Q: My appointment is still "Pending." > Waiting for doctor review. You'll receive a notification when confirmed.

Q: I can't join the video meeting. > Make sure you: (1) accepted the meeting agreement, (2) allowed camera/mic access, (3) use Chrome, (4) meeting hasn't ended already.

Q: Where can I see my lab results? > Go to **PHR** → **Lab & Imaging** tab. Results appear there once your doctor enters them.

Q: The map doesn't show facilities. > Allow location access in browser. Try increasing the distance filter (10 or 20 km).

Q: How do I change the language? > Go to **Settings** → **Language** and toggle Thai ↔ English.

Q: AI Doctor chat isn't responding. > The AI service may need a moment. Try refreshing and starting a new session.

Q: How do I share health records with a doctor? > Go to **PDPA Consent** and enable sharing for specific data types per doctor.

Q: Guest can't enter the meeting. > Guest must wait in the Lobby for doctor approval. Doctor needs to click "Admit" in the Waiting Room panel.

Q: How do I invite family to my meeting? > Copy the guest join link from the pre-join screen and share it. Guests only need a name — no account required.

Q: Where is the AI SOAP summary? > After doctor approves, it appears in your PHR and Timeline. You'll receive a notification.

Q: Can I revoke my Living Will? > Yes. Go to Living Will page → click "Revoke." The document will be immediately marked as revoked.

Q: How do I see other doctors in the system? > Go to **Doctors Directory** in the sidebar. You can search by name or specialty.

Q: What is CDS (Clinical Decision Support)? > AI automatically checks drug interactions, dosage, and allergy contraindications when prescribing. The doctor sees alerts before finalizing prescriptions.

Q: How is my data protected? > All data is encrypted in transit (HTTPS), passwords are hashed (bcrypt), and you control exactly what data is shared via PDPA consent settings.

Q: Meeting recording failed or is too large. > Maximum recording size is 200MB. For longer meetings, pause/resume recording strategically. The transcript is always available regardless of recording status.

Q: แพทย์ใหม่ลงทะเบียนแล้วแต่เข้าสู่ระบบไม่ได้? > ต้องรอผู้ดูแลระบบอนุมัติก่อน (ดูหัวข้อ 6 กระบวนการลงทะเบียนแพทย์ใหม่)

Appendix: Platform Architecture

Component	URL	Purpose
Patient Portal	izara-patient-portal-dev-testing-*.run.app	Patient-facing web application
Doctor Portal	izara-doctor-portal-dev-testing-*.run.app	Doctor and Admin web application
Meeting Server	izara-meeting-server-dev-testing-*.run.app	Jitsi video meeting management
Database	PostgreSQL (Cloud SQL)	All data storage (PHR, EMR, appointments)
AI Engine	Gemini 2.5 Flash	Medical AI features

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This user guide covers v1.5.10 with 2,013 unit tests + 1,149 E2E + 516 UI tests (638 screenshots) from 105 test files.